# Formulari de química general

## TAULA PERIÒDICA DELS ELEMENTS

Nombre atòmic
Símbol
Nom
Configuració electrònica
Potencial d'ionització (en eV)

Pes atòmic
(Els valors entre parèntesis es refereixen a l'isòtop més estable)
Punt de fusió (en °C)
Punt d'ebullició (en °C)
Densitat (en g/mL a 20 °C)
(Elements gasosos, en g/L a 0 °C i 1 atm)
Nombre d'oxidació
Electronegativitat
(Escala de Pauling)

Gas | Líquid (a 30 °C) | Sòlid | Sintètic

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1<br>1,0080<br>−259,34<br>−252,87<br>0,09<br>**H**<br>HIDROGEN<br>+1 −1<br>$1s^1$<br>13,60<br>2,20 | | | | | | | | | | | | | | | | | 2<br>4,0026<br>−272,2<br>−268,9<br>0,18<br>**He**<br>HELI<br>$1s^2$<br>24,59 |
| 2 | 3<br>6,94<br>180,54<br>1342<br>0,53<br>**Li**<br>LITI<br>+1<br>[He] $2s^1$<br>5,39<br>0,98 | 4<br>9,0122<br>1287<br>2471<br>1,85<br>**Be**<br>BERIL·LI<br>+2<br>[He] $2s^2$<br>9,32<br>1,57 | | | | | | | | | | | 5<br>10,81<br>2075<br>4000<br>2,34<br>**B**<br>BOR<br>+3<br>[He] $2s^2 2p^1$<br>8,30<br>2,04 | 6<br>12,011<br>(diamant) 3550<br>4492<br>3,15<br>**C**<br>CARBONI<br>+2 +4 −4<br>[He] $2s^2 2p^2$<br>11,26<br>2,55 | 7<br>14,007<br>−210,00<br>−195,80<br>1,25<br>**N**<br>NITROGEN<br>+3 +5 −3<br>[He] $2s^2 2p^3$<br>14,53<br>3,04 | 8<br>15,999<br>−218,79<br>−182,95<br>1,43<br>**O**<br>OXIGEN<br>−2<br>[He] $2s^2 2p^4$<br>13,62<br>3,44 | 9<br>18,998<br>−219,62<br>−188,12<br>1,69<br>**F**<br>FLUOR<br>−1<br>[He] $2s^2 2p^5$<br>17,42<br>3,98 | 10<br>20,180<br>−248,59<br>−246,08<br>0,90<br>**Ne**<br>NEÓ<br>[He] $2s^2 2p^6$<br>21,56 |
| 3 | 11<br>22,990<br>97,72<br>883<br>0,97<br>**Na**<br>SODI<br>+1<br>[Ne] $3s^1$<br>5,14<br>0,93 | 12<br>24,305<br>650<br>1090<br>1,74<br>**Mg**<br>MAGNESI<br>+2<br>[Ne] $3s^2$<br>7,65<br>1,31 | | | | | | | | | | | 13<br>26,982<br>660,32<br>2519<br>2,70<br>**Al**<br>ALUMINI<br>+3<br>[Ne] $3s^2 3p^1$<br>5,99<br>1,61 | 14<br>28,085<br>1414<br>3265<br>2,33<br>**Si**<br>SILICI<br>+4 −4<br>[Ne] $3s^2 3p^2$<br>8,15<br>1,90 | 15<br>30,974<br>44,15<br>280,4<br>1,82<br>**P**<br>FÒSFOR<br>+3 +5 −3<br>[Ne] $3s^2 3p^3$<br>10,49<br>2,19 | 16<br>32,06<br>115,21<br>444,60<br>2,07<br>**S**<br>SOFRE<br>+4 +6 −2<br>[Ne] $3s^2 3p^4$<br>10,36<br>2,58 | 17<br>35,45<br>−101,5<br>−34,04<br>3,21<br>**Cl**<br>CLOR<br>+1 +3 +5 +7 −1<br>[Ne] $3s^2 3p^5$<br>12,97<br>3,16 | 18<br>39,95<br>−189,3<br>−185,9<br>1,78<br>**Ar**<br>ARGÓ<br>[Ne] $3s^2 3p^6$<br>15,76 |
| 4 | 19<br>39,098<br>63,28<br>759<br>0,86<br>**K**<br>POTASSI<br>+1<br>[Ar] $4s^1$<br>4,34<br>0,82 | 20<br>40,078<br>842<br>1484<br>1,55<br>**Ca**<br>CALCI<br>+2<br>[Ar] $4s^2$<br>6,11<br>1,00 | 21<br>44,956<br>1541<br>2830<br>2,99<br>**Sc**<br>ESCANDI<br>+3<br>[Ar] $3d^1 4s^2$<br>6,56<br>1,36 | 22<br>47,867<br>1668<br>3287<br>4,54<br>**Ti**<br>TITANI<br>+3 +4<br>[Ar] $3d^2 4s^2$<br>6,83<br>1,54 | 23<br>50,942<br>1910<br>3407<br>6,11<br>**V**<br>VANADI<br>+2 +3 +4 +5<br>[Ar] $3d^3 4s^2$<br>6,75<br>1,63 | 24<br>51,996<br>1907<br>2671<br>7,19<br>**Cr**<br>CROM<br>+2 +3 +6<br>[Ar] $3d^5 4s^1$<br>6,77<br>1,66 | 25<br>54,938<br>1246<br>2061<br>7,43<br>**Mn**<br>MANGANÈS<br>+2 +3 +4 +6 +7<br>[Ar] $3d^5 4s^2$<br>7,43<br>1,55 | 26<br>55,845<br>1538<br>2861<br>7,87<br>**Fe**<br>FERRO<br>+2 +3<br>[Ar] $3d^6 4s^2$<br>7,90<br>1,83 | 27<br>58,933<br>1495<br>2927<br>8,9<br>**Co**<br>COBALT<br>+2 +3<br>[Ar] $3d^7 4s^2$<br>7,88<br>1,88 | 28<br>58,693<br>1455<br>2913<br>8,90<br>**Ni**<br>NÍQUEL<br>+2 +3<br>[Ar] $3d^8 4s^2$<br>7,64<br>1,91 | 29<br>63,546<br>1084<br>2562<br>8,96<br>**Cu**<br>COURE<br>+1 +2<br>[Ar] $3d^{10} 4s^1$<br>7,73<br>1,90 | 30<br>65,38<br>419,53<br>907<br>7,13<br>**Zn**<br>ZINC<br>+2<br>[Ar] $3d^{10} 4s^2$<br>9,39<br>1,65 | 31<br>69,723<br>29,76<br>2204<br>5,91<br>**Ga**<br>GAL·LI<br>+3<br>[Ar] $3d^{10} 4s^2 4p^1$<br>6,00<br>1,81 | 32<br>72,630<br>938,25<br>2833<br>5,32<br>**Ge**<br>GERMANI<br>+2 +4<br>[Ar] $3d^{10} 4s^2 4p^2$<br>7,90<br>2,01 | 33<br>74,922<br>(sub.) 614<br>5,73<br>**As**<br>ARSÈNIC<br>+3 +5 −3<br>[Ar] $3d^{10} 4s^2 4p^3$<br>9,79<br>2,18 | 34<br>78,971<br>221<br>685<br>4,79<br>**Se**<br>SELENI<br>+4 +6 −2<br>[Ar] $3d^{10} 4s^2 4p^4$<br>9,75<br>2,55 | 35<br>79,904<br>−7,2<br>58,78<br>3,12<br>**Br**<br>BROM<br>+1 +3 +5 −1<br>[Ar] $3d^{10} 4s^2 4p^5$<br>11,81<br>2,96 | 36<br>83,798<br>−157,36<br>−153,22<br>3,75<br>**Kr**<br>CRIPTÓ<br>[Ar] $3d^{10} 4s^2 4p^6$<br>14,00<br>3,00 |
| 5 | 37<br>85,468<br>39,31<br>688<br>1,53<br>**Rb**<br>RUBIDI<br>+1<br>[Kr] $5s^1$<br>4,18<br>0,82 | 38<br>87,62<br>777<br>1382<br>2,54<br>**Sr**<br>ESTRONCI<br>+2<br>[Kr] $5s^2$<br>5,69<br>0,95 | 39<br>88,906<br>1526<br>3336<br>4,47<br>**Y**<br>ITRI<br>+3<br>[Kr] $4d^1 5s^2$<br>6,22<br>1,22 | 40<br>91,224<br>1855<br>4409<br>6,51<br>**Zr**<br>ZIRCONI<br>+4<br>[Kr] $4d^2 5s^2$<br>6,63<br>1,33 | 41<br>92,906<br>2477<br>4744<br>8,57<br>**Nb**<br>NIOBI<br>+3 +5<br>[Kr] $4d^4 5s^1$<br>6,76<br>1,60 | 42<br>95,95<br>2623<br>4639<br>10,22<br>**Mo**<br>MOLIBDÈ<br>+4 +5 +6<br>[Kr] $4d^5 5s^1$<br>7,09<br>2,16 | 43<br>(97)<br>2157<br>4265<br>11,5<br>**Tc**<br>TECNECI<br>+6 +7<br>[Kr] $4d^5 5s^2$<br>7,12<br>1,90 | 44<br>101,07<br>2334<br>4150<br>12,41<br>**Ru**<br>RUTENI<br>+3 +4 +6<br>[Kr] $4d^7 5s^1$<br>7,36<br>2,20 | 45<br>102,91<br>1964<br>3695<br>12,41<br>**Rh**<br>RODI<br>+1 +3 +4<br>[Kr] $4d^8 5s^1$<br>7,46<br>2,28 | 46<br>106,42<br>1554,9<br>2963<br>12,02<br>**Pd**<br>PAL·LADI<br>+2 +4<br>[Kr] $4d^{10}$<br>8,34<br>2,20 | 47<br>107,87<br>961,78<br>2162<br>10,50<br>**Ag**<br>PLATA<br>+1<br>[Kr] $4d^{10} 5s^1$<br>7,58<br>1,93 | 48<br>112,41<br>321,07<br>767<br>8,65<br>**Cd**<br>CADMI<br>+2<br>[Kr] $4d^{10} 5s^2$<br>8,99<br>1,69 | 49<br>114,82<br>156,60<br>2072<br>7,31<br>**In**<br>INDI<br>+3<br>[Kr] $4d^{10} 5s^2 5p^1$<br>5,79<br>1,78 | 50<br>118,71<br>231,9<br>2602<br>5,75<br>**Sn**<br>ESTANY<br>+2 +4<br>[Kr] $4d^{10} 5s^2 5p^2$<br>7,34<br>1,96 | 51<br>121,76<br>630,63<br>1587<br>6,69<br>**Sb**<br>ANTIMONI<br>+3 +5 −3<br>[Kr] $4d^{10} 5s^2 5p^3$<br>8,61<br>2,05 | 52<br>127,60<br>449,51<br>988<br>6,24<br>**Te**<br>TEL·LURI<br>+4 +6 −2<br>[Kr] $4d^{10} 5s^2 5p^4$<br>9,01<br>2,10 | 53<br>126,90<br>113,7<br>184,4<br>4,93<br>**I**<br>IODE<br>+1 +5 +7 −1<br>[Kr] $4d^{10} 5s^2 5p^5$<br>10,45<br>2,66 | 54<br>131,29<br>−111,75<br>−108<br>5,90<br>**Xe**<br>XENÓ<br>+2 +4 +6<br>[Kr] $4d^{10} 5s^2 5p^6$<br>12,13<br>2,60 |
| 6 | 55<br>132,91<br>28,44<br>671<br>1,87<br>**Cs**<br>CESI<br>+1<br>[Xe] $6s^1$<br>3,89<br>0,79 | 56<br>137,33<br>727<br>1897<br>3,5<br>**Ba**<br>BARI<br>+2<br>[Xe] $6s^2$<br>5,21<br>0,89 | 57<br>71<br>**La-Lu** | 72<br>178,49<br>2233<br>4603<br>13,31<br>**Hf**<br>HAFNI<br>+4<br>[Xe] $4f^{14} 5d^2 6s^2$<br>6,83<br>1,30 | 73<br>180,95<br>3017<br>5458<br>16,65<br>**Ta**<br>TÀNTAL<br>+5<br>[Xe] $4f^{14} 5d^3 6s^2$<br>7,55<br>1,50 | 74<br>183,84<br>3422<br>5655<br>19,3<br>**W**<br>TUNGSTÈ<br>+4 +5 +6<br>[Xe] $4f^{14} 5d^4 6s^2$<br>7,86<br>2,36 | 75<br>186,21<br>3186<br>5596<br>21,02<br>**Re**<br>RENI<br>+4 +6 +7<br>[Xe] $4f^{14} 5d^5 6s^2$<br>7,83<br>1,90 | 76<br>190,23<br>3033<br>5012<br>22,57<br>**Os**<br>OSMI<br>+4 +6 +8<br>[Xe] $4f^{14} 5d^6 6s^2$<br>8,44<br>2,20 | 77<br>192,22<br>2446<br>4428<br>22,42<br>**Ir**<br>IRIDI<br>+1 +3 +4<br>[Xe] $4f^{14} 5d^7 6s^2$<br>8,97<br>2,20 | 78<br>195,08<br>1768,4<br>3825<br>21,45<br>**Pt**<br>PLATÍ<br>+2 +4<br>[Xe] $4f^{14} 5d^9 6s^1$<br>8,96<br>2,28 | 79<br>196,97<br>1064,18<br>2856<br>19,3<br>**Au**<br>OR<br>+1 +3<br>[Xe] $4f^{14} 5d^{10} 6s^1$<br>9,23<br>2,54 | 80<br>200,59<br>−38,83<br>356,73<br>13,55<br>**Hg**<br>MERCURI<br>+1 +2<br>[Xe] $4f^{14} 5d^{10} 6s^2$<br>10,44<br>2,00 | 81<br>204,38<br>304<br>1473<br>11,85<br>**Tl**<br>TAL·LI<br>+1 +3<br>[Xe] $4f^{14} 5d^{10} 6s^2 6p^1$<br>6,11<br>1,62 | 82<br>207,2<br>327,46<br>1749<br>11,35<br>**Pb**<br>PLOM<br>+2 +4<br>[Xe] $4f^{14} 5d^{10} 6s^2 6p^2$<br>7,42<br>2,33 | 83<br>208,98<br>271,4<br>1564<br>9,75<br>**Bi**<br>BISMUT<br>+3 +5<br>[Xe] $4f^{14} 5d^{10} 6s^2 6p^3$<br>7,29<br>2,02 | 84<br>(209)<br>254<br>962<br>9,32<br>**Po**<br>POLONI<br>+2 +4<br>[Xe] $4f^{14} 5d^{10} 6s^2 6p^4$<br>8,41<br>2,00 | 85<br>(210)<br>302,33<br>7<br>**At**<br>ÀSTAT<br>+1 −1<br>[Xe] $4f^{14} 5d^{10} 6s^2 6p^5$<br>9,32<br>2,20 | 86<br>(222)<br>−71<br>−61,7<br>**Rn**<br>RADÓ<br>[Xe] $4f^{14} 5d^{10} 6s^2 6p^6$<br>10,75 |
| 7 | 87<br>(223)<br>27<br>677<br>**Fr**<br>FRANCI<br>+1<br>[Rn] $7s^1$<br>4,07<br>0,70 | 88<br>(226)<br>700<br>1140<br>5,0<br>**Ra**<br>RADI<br>+2<br>[Rn] $7s^2$<br>5,28<br>0,90 | 89<br>103<br>**Ac-Lr** | 104<br>(267)<br>**Rf**<br>RUTHERFORDI<br>[Rn] $5f^{14} 6d^2 7s^2$<br>6,01 | 105<br>(268)<br>**Db**<br>DUBNI<br>[Rn] $5f^{14} 6d^3 7s^2$<br>6,8 | 106<br>(269)<br>**Sg**<br>SEABORGI<br>[Rn] $5f^{14} 6d^4 7s^2$<br>7,8 | 107<br>(278)<br>**Bh**<br>BOHRI<br>[Rn] $5f^{14} 6d^5 7s^2$<br>7,7 | 108<br>(278)<br>**Hs**<br>HASSI<br>[Rn] $5f^{14} 6d^6 7s^2$<br>7,6 | 109<br>(282)<br>**Mt**<br>MEITNERI | 110<br>(282)<br>**Ds**<br>DARMSTADTI | 111<br>(286)<br>**Rg**<br>ROENTGENI | 112<br>(286)<br>**Cn**<br>COPERNICI | 113<br>(286)<br>**Nh**<br>NIHONI | 114<br>(290)<br>**Fl**<br>FLEROVI | 115<br>(290)<br>**Mc**<br>MOSCOVI | 116<br>(293)<br>**Lv**<br>LIVERMORI | 117<br>(294)<br>**Ts**<br>TENNES | 118<br>(294)<br>**Og**<br>OGANESSÓ |

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 6 | 57<br>138,91<br>920<br>3455<br>6,14<br>**La**<br>LANTANI<br>+3<br>[Xe] $5d^1 6s^2$<br>5,58<br>1,10 | 58<br>140,12<br>799<br>3424<br>6,77<br>**Ce**<br>CERI<br>+3 +4<br>[Xe] $4f^1 5d^1 6s^2$<br>5,54<br>1,12 | 59<br>140,91<br>931<br>3510<br>6,77<br>**Pr**<br>PRASEODIMI<br>+3<br>[Xe] $4f^3 6s^2$<br>5,47<br>1,13 | 60<br>144,24<br>1016<br>3066<br>7,01<br>**Nd**<br>NEODIMI<br>+3<br>[Xe] $4f^4 6s^2$<br>5,52<br>1,14 | 61<br>(145)<br>1042<br>−3000<br>7,26<br>**Pm**<br>PROMETI<br>+3<br>[Xe] $4f^5 6s^2$<br>5,58 | 62<br>150,36<br>1072<br>1790<br>7,52<br>**Sm**<br>SAMARI<br>+2 +3<br>[Xe] $4f^6 6s^2$<br>5,64<br>1,17 | 63<br>151,96<br>822<br>1596<br>5,24<br>**Eu**<br>EUROPI<br>+2 +3<br>[Xe] $4f^7 6s^2$<br>5,67 | 64<br>157,25<br>1314<br>3264<br>7,90<br>**Gd**<br>GADOLINI<br>+3<br>[Xe] $4f^7 5d^1 6s^2$<br>6,15<br>1,20 | 65<br>158,93<br>1359<br>3221<br>8,23<br>**Tb**<br>TERBI<br>+3<br>[Xe] $4f^9 6s^2$<br>5,86 | 66<br>162,50<br>1411<br>2561<br>8,55<br>**Dy**<br>DISPROSI<br>+3<br>[Xe] $4f^{10} 6s^2$<br>5,94<br>1,22 | 67<br>164,93<br>1472<br>2694<br>8,80<br>**Ho**<br>HOLMI<br>+3<br>[Xe] $4f^{11} 6s^2$<br>6,02<br>1,23 | 68<br>167,26<br>1529<br>2862<br>9,06<br>**Er**<br>ERBI<br>+3<br>[Xe] $4f^{12} 6s^2$<br>6,11<br>1,24 | 69<br>168,93<br>1545<br>1946<br>9,32<br>**Tm**<br>TULI<br>+3<br>[Xe] $4f^{13} 6s^2$<br>6,18<br>1,25 | 70<br>173,05<br>824<br>1194<br>6,90<br>**Yb**<br>ITERBI<br>+2 +3<br>[Xe] $4f^{14} 6s^2$<br>6,25 | 71<br>174,97<br>1663<br>3393<br>9,84<br>**Lu**<br>LUTECI<br>+3<br>[Xe] $4f^{14} 5d^1 6s^2$<br>5,43<br>1,27 |
| 7 | 89<br>(227)<br>1051<br>3200<br>**Ac**<br>ACTINI<br>+3<br>[Rn] $6d^1 7s^2$<br>5,38<br>1,10 | 90<br>232,04<br>1750<br>4788<br>11,72<br>**Th**<br>TORI<br>+4<br>[Rn] $6d^2 7s^2$<br>6,31<br>1,30 | 91<br>231,04<br>1572<br>15,37<br>**Pa**<br>PROTOACTINI<br>+3 +4 +5<br>[Rn] $5f^2 6d^1 7s^2$<br>5,89<br>1,50 | 92<br>238,03<br>1135<br>4131<br>19,95<br>**U**<br>URANI<br>+3 +4 +5 +6<br>[Rn] $5f^3 6d^1 7s^2$<br>6,19<br>1,38 | 93<br>(237)<br>644<br>3902<br>20,25<br>**Np**<br>NEPTUNI<br>+3 +4 +5 +6<br>[Rn] $5f^4 6d^1 7s^2$<br>6,27<br>1,36 | 94<br>(244)<br>640<br>3228<br>19,84<br>**Pu**<br>PLUTONI<br>+3 +4 +5 +6<br>[Rn] $5f^6 7s^2$<br>6,03<br>1,28 | 95<br>(243)<br>1176<br>2607<br>13,67<br>**Am**<br>AMERICI<br>+3 +4 +5 +6<br>[Rn] $5f^7 7s^2$<br>5,97<br>1,30 | 96<br>(247)<br>1345<br>13,51<br>**Cm**<br>CURI<br>+3<br>[Rn] $5f^7 6d^1 7s^2$<br>5,99<br>1,30 | 97<br>(247)<br>**Bk**<br>BERKELI<br>+3 +4<br>[Rn] $5f^9 7s^2$<br>6,20<br>1,30 | 98<br>(251)<br>**Cf**<br>CALIFORNI<br>+3<br>[Rn] $5f^{10} 7s^2$<br>6,28<br>1,30 | 99<br>(252)<br>**Es**<br>EINSTEINI<br>+3<br>[Rn] $5f^{11} 7s^2$<br>6,37<br>1,30 | 100<br>(257)<br>**Fm**<br>FERMI<br>+3<br>[Rn] $5f^{12} 7s^2$<br>6,50<br>1,30 | 101<br>(258)<br>**Md**<br>MENDELEVI<br>+3<br>[Rn] $5f^{13} 7s^2$<br>6,58<br>1,30 | 102<br>(259)<br>**No**<br>NOBELI<br>+3<br>[Rn] $5f^{14} 7s^2$<br>6,65<br>1,30 | 103<br>(266)<br>**Lr**<br>LAWRENCI<br>+3<br>[Rn] $5f^{14} 7s^2 7p^1$<br>4,96 |

SCQ Societat Catalana de Química

## Fórmules

| Fórmula | Descripció |
|---|---|
| $p = mv$ | Moment lineal, la massa i la velocitat |
| $P = \frac{F}{A}$ | Definició de pressió |
| $PV = n\mathrm{R}T$ | Llei dels gasos ideals |
| $PV = \frac{2}{3}\langle E_c\rangle = \frac{2}{3}N_\mathrm{A}\left\langle\frac{mc^2}{2}\right\rangle$ | Fórmula de la teoria cinètica dels gasos |
| $\langle E_c\rangle = N_\mathrm{A}\frac{m\langle c^2\rangle}{2} = \frac{3}{2}\mathrm{R}T$ | Energia cinètica translacional d'un mol de gas ideal |
| $\frac{\Delta N}{N} = 4\pi\left(\frac{m}{2\pi\mathrm{k}T}\right)^{3/2}e^{-mc^2/2\mathrm{k}T}c^2\Delta c$ | Distribució de Maxwell-Boltzmann |
| $z = \frac{V_m}{V_{m,i}} = \frac{V_m}{\mathrm{R}T/P} = \frac{PV_m}{\mathrm{R}T}$ | Factor de compressibilitat dels gasos no ideals |
| $\left(P + \frac{n^2a}{V^2}\right)(V - nb) = n\mathrm{R}T$ | Equació de van der Waals |
| $w = -P\Delta V$ | Treball exercit sobre un gas |
| $\Delta U = q + w$ | Primera llei de la termodinàmica |
| $H = U + PV$ | Entalpia |
| $\mathrm{d}S = \frac{\mathrm{d}q_\mathrm{rev}}{T}$ | Definició d'entropia |
| $G = H - TS$ | Energia lliure de Gibbs |
| $q_v = n\Delta U$ | Calor a volum constant |
| $q_p = n\Delta H$ | Calor a pressió constant |
| $C_v = \left(\frac{\partial U}{\partial T}\right)_V = \left(\frac{\partial q_v}{\partial T}\right)_V$ | Capacitat calorífica a volum constant |
| $C_p = \left(\frac{\partial H}{\partial T}\right)_P = \left(\frac{\partial q_p}{\partial T}\right)_P$ | Capacitat calorífica a pressió constant |
| $C_p - C_v = \mathrm{R}$ | Relació entre capacitats calorífiques d'un gas ideal |
| $\Delta G = \Delta H - T\Delta S$ | Canvi d'energia lliure de Gibbs |
| $\Delta G^\circ = -\mathrm{R}T\ln K$ | Energia lliure de Gibbs i constant d'equilibri |
| $E^\circ_\text{pila} = E^\circ_\text{càtode} - E^\circ_\text{ànode}$ | Potencial estàndard de la pila |
| $E = E^\circ - \frac{RT}{n\mathrm{F}}\ln Q$ | Equació de Nernst, f.e.m. (E) i quocient de reacció |
| $\Delta G = -n\mathrm{F}E$ | Energia lliure de Gibbs i potencial elèctric |
| $K = Q_\mathrm{eq} = \frac{\prod_i^P[\text{productes}_i]^{\text{coef}_i}}{\prod_j^R[\text{reactius}_j]^{\text{coef}_j}}$ | Constant d'equilibri |
| $K_p = K_c(\mathrm{R}T)^{\Delta n}$ | Constant d'equilibri (pressió-concentració) |
| $K_{sp} = \prod_i^{ions}[\text{ions}_i]^{\text{coef}_i}$ | Producte de solubilitat |
| $K_a = \frac{[\mathrm{H}^+][\mathrm{A}^-]}{[\mathrm{HA}]}$ | Constant d'acidesa |
| $K_b = \frac{[\mathrm{OH}^-][\mathrm{BH}^+]}{[\mathrm{B}]}$ | Constant de basicitat |
| $pK_a = -\log K_a$ | Constant d'acidesa i pKa |
| $pH = -\log[\mathrm{H}^+]$ | Definició de pH |
| $K_w = [\mathrm{H}^+][\mathrm{OH}^-] = K_a \cdot K_b$ | Producte iònic de l'aigua |
| $pK_w = pH + pOH$ | pH, pOH i pKw |
| $\mathrm{pH} = \mathrm{pKa} + \log\frac{[\mathrm{A}^-]}{[\mathrm{HA}]}$ | Equació de Henderson-Hasselbalch |
| $C_i = k_H \cdot P_i$ | Llei de Henry |
| $C = \frac{n}{V}$ | Concentració molar |
| $P_A = X_A P_A^\circ$ | Llei de Raoult: pressió parcial component en solució |
| $X_A = \frac{n_A}{n_A + n_B}$ | Fracció molar component solució |
| $m = \frac{n}{m_\text{solvent}}$ | Definició de molalitat |
| $\Delta T_b = K_b \cdot m$ | Elevació del punt d'ebullició |
| $\Delta T_f = K_f \cdot m$ | Descens del punt de congelació |
| $V = IR$ | Llei d'Ohm |
| $P = IV$ | Potència elèctrica |
| $\Delta E = P\Delta t$ | Energia elèctrica |
| $C_\text{cal} = \frac{q_\text{absorvit}}{\Delta T}$ | Factor de calibratge del calorímetre (capacitat calorífica) |
| $\Delta E = \mathrm{h}\nu$ | Energia d'un fotó |
| $\lambda = \frac{\mathrm{h}}{mv}$ | Longitud d'ona de de Broglie |
| $\Delta E = \frac{\mathrm{h}c}{\lambda}$ | Relació entre energia, constant de Planck i longitud d'ona |
| $E_n = -\frac{Z^2 m_e e^4}{8\varepsilon_0^2\mathrm{h}^2 n^2}$ | Energia de l'òrbita $n$ (model de Bohr) |
| $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$ | Principi d'incertesa de Heisenberg |

## Constants

| Constant | Valor |
|---|---|
| Constant dels gasos | $0{,}082\,\mathrm{atm\,L\,mol^{-1}\,K^{-1}} =$ $8{,}314\,5\,\mathrm{J\,K^{-1}\,mol^{-1}} = 8{,}314\,5\,\mathrm{kPa\,L\,K^{-1}\,mol^{-1}}$ |
| Número/Constant d'Avogadro ($N_\mathrm{A}$) | $6{,}022 \cdot 10^{23}\,\mathrm{mol^{-1}}$ |
| Càrrega d'un electró | $1{,}602 \cdot 10^{-19}\,\mathrm{C}$ |
| Massa d'un electró | $9{,}109 \cdot 10^{-31}\,\mathrm{kg}$ |
| Massa d'un protó | $1{,}673 \cdot 10^{-27}\,\mathrm{kg}$ |
| Massa d'un neutró | $1{,}675 \cdot 10^{-27}\,\mathrm{kg}$ |
| Constant de Planck (h) | $6{,}626 \cdot 10^{-34}\,\mathrm{J\,s}$ |
| $\hbar = \frac{\mathrm{h}}{2\pi}$ (constant reduïda de Planck) | $1{,}055 \cdot 10^{-34}\,\mathrm{J\,s}$ |
| Constant de Boltzmann | $1{,}381 \cdot 10^{-23}\,\mathrm{J\,K^{-1}}$ |
| Relació entre R i $\mathrm{k_B}$ | $\mathrm{R} = N_\mathrm{A}\mathrm{k_B}$ |
| Constant de Faraday | $96\,485\,\mathrm{C\,mol^{-1}}$ |
| Constant de gravitació universal | $6{,}674 \cdot 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ |
| Constant de Coulomb | $8{,}988 \cdot 10^{9}\,\mathrm{N\,m^2\,C^{-2}}$ |
| Constant de Rydberg | $1{,}097 \cdot 10^{7}\,\mathrm{m^{-1}}$ |

## Unitats de mesura

| Magnitud | Unitat a SI | Símbol SI | Dimensió |
|---|---|---|---|
| Longitud | metre | m | $\mathsf{L}$ |
| Volum | litre | L | $\mathsf{L}^3$ |
| Massa | kilogram | kg | $\mathsf{M}$ |
| Temperatura | kelvin | K | |
| mol | mol | mol | $\mathsf{N}$ |
| temps | segon | s | $\mathsf{T}$ |
| Freqüència | hertz | Hz | $\mathsf{T}^{-1}$ |
| Energia | joule | J | $\mathsf{ML^2T^{-2}}$ |
| Força | newton | N | $\mathsf{MLT^{-2}}$ |
| Pressió | pascal | Pa | $\mathsf{ML^{-1}T^{-2}}$ |
| Potencial elèctric | volt | V | $\mathsf{ML^2T^{-3}I^{-1}}$ |
| Potència | watt | W | $\mathsf{ML^2T^{-3}}$ |

| Magnitud | Unitat (EUA) | Equivalència en SI |
|---|---|---|
| Volum | $1\,\mathrm{in}^3$ | $16{,}387\,\mathrm{cm}^3$ |
| Volum | $1\,\mathrm{ft}^3$ | $28{,}317\,\mathrm{L}$ |
| Volum | 1 gal (US) | $3{,}785\,\mathrm{L}$ |
| Pressió | 1 psi | $6{,}895\,\mathrm{kPa}$ |
| Pressió | 1 atm | $101{,}325\,\mathrm{kPa}$ |
| Pressió | 1 inHg | $3{,}386\,\mathrm{kPa}$ |
| Temperatura | $1\,\mathrm{F}$ | $T_C = (T_F - 32) \times \frac{5}{9}$ |
| Massa | 1 oz | $28{,}35\,\mathrm{g}$ |
| Massa | 1 lb | $0{,}453\,6\,\mathrm{kg}$ |
| Massa | 1 t (US) | $907{,}184\,\mathrm{kg}$ |

| Unitat de Pressió | Pressió (en relació a 1 atm) |
|---|---|
| Atmosfera (atm) | 1 atm |
| Pascal (Pa) | 101 325 Pa |
| Kilopascal (kPa) | 101,325 kPa |
| Bar | 1,013 25 bar |
| Mil·límetre de mercuri (mmHg) | 760 Torr |
| Torra (Torr) | 760 Torr |
| Pounds per square inch (psi) | 14,696 psi |

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